

A Reflective Journey: Navigating Your Cumulative Experience at Iowa State University

By Randolph Roman

Introduction

The past four years at Iowa State have been a very influential time in my life. I have gained knowledge and life experiences that I might never have dreamed of if not for my time here. I have drastically increased my foundation of knowledge and become better prepared to enter the workforce, as well as making connections with classmates who might someday be colleagues. I have both intellectually and culturally broadened my horizons through the study abroad program here, immersing myself in the norms there, and connecting with people.

As for preparation for my future career, Iowa state has provided me with an immeasurable number of resources, spanning from knowledge acquired in classes, to connections with professors as well as peers, and everything in between. Though there were challenges to face throughout my time here, I have emerged from those struggles even more resilient than before, with a newfound eagerness to approach situations with a more optimistic attitude, as well as new skills under my belt. Throughout the entirety of my time here, there has never been a moment where I regretted my decision to attend Iowa State University.

Embracing the Bigger Picture

While attending Iowa State University, I had the opportunity to experience its diverse education which has prepared me for any challenges that come my way. ISU offers a wide variety of classes to take, some required and others chosen by each student to suit their needs and help them pursue their individual interests. Classes like CPRE 3810 where students design a basic CPU for computing assembly code taught me how to take steps to design intricate computing systems and collaborate with a team. My senior design project has taught me even more on teamwork, from how I work in a larger group and how to divide up work to taking accountability for mistakes to how to ask for help when I need assistance on completing a deliverable.

I've had multiple classes that give presentations, deepening and strengthening my public speaking abilities. Some even helped me handle complex modern issues, like the topic of AI. Multiple classes handle this topic in different ways, with some encouraging it, curious with this integration into a classroom and learning environment, with others limiting it in certain

situations, changing grading criteria to accommodate, from even outright banning the technology. These different approaches have granted me with different perspectives into this complex issue, making me navigate this ethical dilemma in a meaningful way and how to approach similar issues.

While I typically haven't gone out to seek external sources like the library or professional journals, I have gone out of my way to deepen my knowledge through both talking with peers or informational videos on the internet like YouTube. One such example that really stuck with me was when I was learning about linear algebra. This was my first math class that felt unique and didn't just feel like the path I had been taking the last few years on the calculus trajectory. This class made me love math again, and as a result I wanted to learn more. So, there was a brief talk about rotation matrices, but it was never something we discussed in class. I wanted to learn more about them. I went home and watched a couple different videos about that, including their importance in computer graphics which is another thing I was very interested in, and really broadening my learning and understanding of this topic outside of the classroom environment. This knowledge later helped me when I took an introduction to computer graphics class, and it was a really cool moment for me when I realized that the learning I did outside the classroom was beneficial and being applied practically.

Embracing Lifelong Experiences Outside of the Classroom

There are some moments that, while still because of ISU, happened outside of an academic setting that really helped me grow and create lasting memories and bonds that I will cherish forever. The first of these experiences I would like to talk about is my involvement of the gymnastics club. I was part of my high school's gymnastic team, so I was very pleased and happy to find out there was a club here I could join and be a part of. I kept to myself most of the time, apart from my friend from high school who was also here and did gymnastics in high school. As I continued to attend, I got more involved, making friends with everyone on the club, going to the meets as a team to different states, and finally being elected two leadership roles: Risk and Safety Management my junior year and this year being the vice president of the club. These roles both taught me a lot about being a leader and having that responsibility. From filling out forms, keeping people accountable for waivers signed, to even setting up and making trip requests for our meets. While the leadership roles taught me a lot, what I will remember most about this club are the friends and memories I've made.

A truly life changing experience I've had here, however, was through the study abroad program to go to Japan offered here through English 3140. This experience was truly one of a kind, inspiring, and the bonds with the people I made on this trip are lifelong. We are all in a group-chat and still meet up regularly, some of us are even planning on going back. I learned so much about Japan and the culture there that I wouldn't have been able to get just learning

about it from books or the internet. Getting to experience it was such a special experience, and all of us in that group sharing that experience made us grow and connect so much more quickly and stronger. I learned so much about myself while overseas, and it genuinely changed and shaped who I am today. This class was about technical communication, and I did learn how to do that properly and professionally, considering things like the barriers between cultures. Through this class and the trip, however, it also taught me how to communicate better as a whole. I became less anxious about public speaking or speaking to strangers, I learned how to effectively communicate my thoughts. It also helped me with time management and better money management. This was such a special class and experience for me and the amount of growth and self-reflection I did as a person is hard to describe, but it is something I will always be grateful for about Iowa State, because I would have never been able to have that experience without it.

Pioneering Growth through Adaptation

My growth has not stopped, however, as I am constantly learning and trying to improve wherever I can. Earlier in the second semester of my senior design project, I was tasked with creating this hardware challenge that I had never heard of before. Thanks to what I've learned here at ISU, I was able to adapt to this change and learned how to complete this challenge. The skills I've learned go beyond just what was taught to me in the classroom but allow me to apply these skills to learning when faced with a new problem or challenge in the real world. I'm able to adapt to the situation when needed and I know that no matter what is thrown at me I'll be able to overcome those challenges.

The way I learn and the strategies I use to overcome complex engineering concepts and problems has evolved since coming to ISU. I started out not really implementing any strategies, choosing to brute force my way through a problem mentally, just throwing myself at it until I succeed. As I continued my education, however, I quickly learned that this approach wasn't viable. I wanted to keep the tenacity to push through a challenge and not give up, but I needed to learn how to do this effectively and efficiently to save myself both time and sanity. I started asking peers more, doing more research into it on my own, and fully analyzing the situation before proceeding. This ability to analyze and seek information has greatly deepened my understanding of what I have learned in order to solve these problems. Instead of bashing my head against a wall until I get it to work and understand what I did and why it worked after getting the solution, I now get a deeper understanding of what to do given a specific task which gives me a framework that allows me to solve the problem quicker and fully understand why what I did worked when I reach the solution.

Crafting a Narrative of Growth

If I started out at Iowa State again as a freshman, there are some things I'd choose to do differently. I think the first thing I would do is change some of the courses I took, more specifically when I took them. There were a couple times I tried to take a class and it just wasn't being offered, knowing this now I would change up when I took some class as I feel like this would have helped the transfer of knowledge as well as keep me from being overwhelmed some semesters and overall improve my mental health. Another thing I would probably do is join a club that helps me improve my technical skills. While gymnastics club has been amazing, I also wish I did a club that would require me to use the knowledge I've learned in classes more or maybe do a personal project throughout my years here that I can show to potential employers to really show off my skills. By the time I realized this I was already in the middle of my senior year and due to mental health and stress I couldn't handle joining a club like that. These are the two major changes I would make if I was able to start over, while there are some minor changes I would make as well, these two are the most important in my opinion.

The Path Ahead: Continuous Development

While it's important to look back, reflect on how I've grown and acknowledge mistakes I would change given the knowledge I have now, it's arguably more important to keep looking forward, to take the knowledge I have gained and apply it to the future. Growth and development never stop, nor should it, and there are areas I'd like to improve on within my knowledge base as I continue my journey. One of which is to continue to improve my knowledge of different coding languages. I'm well versed in languages like Java and C, but I am lacking in others such as C++ or Python. While I have some experience with these languages, it's not as extensive as the other languages I mentioned, and as a result, I'd like to learn more about them and become more proficient in them. There are also several algorithms or applications that would be very useful to have a better understanding of. The world of computers and coding is ever evolving, and as a computer engineer, I need to be able to keep up with this ever-evolving landscape.

It's hard to know where I'll end up in a few years, I'm not the best and am planning out or envisioning the future that way because I like to live in the moment and savor it. With that being said, in broad strokes I can imagine myself finding a good starting position within the next year. From this position, I will continue to hone my skills and make good connections. Maybe I'll like this place enough to stay for longer, but I imagine I'd try to find a better position after a couple years of work to continue growing professionally and mentally. I don't know how many times this will happen, but eventually I'll find a place that allows me to continue developing my skills within this job, which will give me the opportunity to never stop growing as a person. I don't know how realistic this plan for my future is, but it doesn't have to be. As long as I keep

growing and developing the skills and techniques I learned here at Iowa State University, I know things will turn out all right.